Microscope



Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **Medium** 3 labs

Covers these lab items: Microscope

What it is used for

A microscope magnifies small objects and structures so they can be examined in greater detail. Correct slide preparation, focusing, and lens care help produce clear images and reliable observations.

Main parts

- Eyepiece or ocular lens
- Objective lenses and revolving nosepiece
- Stage and slide holder
- Coarse and fine focus controls
- Illuminator or mirror and brightness control
- Base, arm, and power switch where applicable

Before you start

1. Place the microscope on a stable, clean surface away from vibration and direct sunlight.
2. Inspect the eyepiece, objective lenses, stage, power cable, and focus controls for damage.
3. Clean lenses only with approved lens paper and suitable lens-cleaning solution.
4. Prepare a correctly labelled slide with a coverslip when required by the specimen method.
5. Start with the lowest-power objective and ensure the stage and lens are clear.
6. Switch on the illuminator and adjust the light level to a suitable starting setting.

How to use

1. Place the slide on the stage and secure it with the slide holder.
2. Centre the specimen over the stage opening using the stage controls.
3. Look through the eyepiece and adjust the light or diaphragm for suitable contrast.
4. Use the coarse focus control carefully to bring the specimen into initial focus at low magnification.
5. Use the fine focus control to sharpen the image.
6. Adjust the illumination and diaphragm to improve image contrast without excessive brightness.
7. Rotate to a higher-power objective only after centring and focusing the specimen at low power.
8. Use fine focus for higher magnification and avoid allowing the objective lens to touch the slide.
9. Record observations and magnification as required by the laboratory procedure.
10. After examination, return to the lowest-power objective, remove the slide, switch off the light, and clean the work area.

⚠ Safety

- Carry the microscope with one hand supporting the arm and the other supporting the base.
- Do not touch glass slides with bare hands when specimens or reagents may be hazardous.
- Keep liquids away from electrical parts and disconnect power before approved maintenance.
- Avoid forcing focus controls or rotating the nosepiece against resistance.
- Dispose of broken slides and contaminated specimens in the designated containers.
- Do not clean lenses with tissues, clothing, or unapproved solvents.

Care & storage

- Clean eyepiece and objective lenses with approved lens paper after use when needed.
- Keep the microscope covered and store it in a dry, dust-free location.
- Inspect the stage, focus controls, illuminator, and power cable regularly.
- Arrange servicing and optical alignment checks according to the manufacturer's recommendations.

Common mistakes to avoid

- Using coarse focus at high magnification can cause the objective to strike the slide.
- A dirty lens or slide can make the image blurry or misleading.
- Excessive illumination can reduce contrast and hide specimen details.
- Carrying the microscope by the eyepiece or stage can damage the instrument.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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